Linear Algebra and Its Applications

Section 6.4: The Gram–Schmidt Process

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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6.4 The Gram–Schmidt process

Gram–Schmidt replaces an independent ordered list by orthogonal directions with the same successive spans. At stage k, subtract the projections of x_k onto every earlier orthogonal direction. Independence guarantees the residual is nonzero; normalizing it gives the next unit direction.

For a full-column-rank matrix A, the normalized columns form Q with $Q^TQ = I$. The factor $R = Q^TA$ is square upper triangular and satisfies $A = QR$. Its diagonal can be made positive by using positive lengths in the normalizations. In the calculations below we verify both $Q^TQ = I$ and $QR = A$, not merely the shape of the proposed factors.

Practice problem 6.4.P1: Normalize a pair that is already orthogonal

The two given vectors satisfy $(1,1,1)\cdot(1/3,1/3,-2/3) = 0$. Their lengths are $\sqrt{3}$ and $\sqrt{6}/3$, respectively. Hence an orthonormal basis is

$$q_1 = \frac{1}{\sqrt{3}}(1,1,1)^T, \qquad q_2 = \frac{1}{\sqrt{6}}(1,1,-2)^T.$$

Each has squared length one, their dot product is zero, and their nonzero rescalings span the same space as the original pair.

Conclusion. The displayed q_1, q_2 are the desired orthonormal basis.

Exercise 6.4.1: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 3 & 8 \\ 0 & 5 \\ -1 & -6 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j}: \quad (3).$$

Subtract all of those components:

$$v_2 = x_2 - (3)w_1 = \begin{bmatrix} -1 \\ 5 \\ -3 \end{bmatrix}.$$

Choose $w_2 = v_2 = \begin{bmatrix} -1 \\ 5 \\ -3 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 10 & 0 \\ 0 & 35 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.2: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 0 & 5 \\ 4 & 6 \\ 2 & -7 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 0 \\ 4 \\ 2 \end{bmatrix}$. Choose $w_1 = (\frac{1}{2})v_1 = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (1).$$

Subtract all of those components:

$$v_2 = x_2 - (1)w_1 = \begin{bmatrix} 5 \\ 4 \\ -8 \end{bmatrix}.$$

Choose $w_2 = v_2 = \begin{bmatrix} 5 \\ 4 \\ -8 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 5 & 0 \\ 0 & 105 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.3: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 2 & 4 \\ -5 & -1 \\ 1 & 2 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (\frac{1}{2}).$$

Subtract all of those components:

$$v_2 = x_2 - (\frac{1}{2})w_1 = \begin{bmatrix} 3 \\ \frac{3}{2} \\ \frac{3}{2} \end{bmatrix}.$$

Choose $w_2 = (\frac{2}{3})v_2 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 30 & 0 \\ 0 & 6 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.4: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 3 & -3 \\ -4 & 14 \\ 5 & -7 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 3 \\ -4 \\ 5 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 3 \\ -4 \\ 5 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (-2).$$

Subtract all of those components:

$$v_2 = x_2 - (-2)w_1 = \begin{bmatrix} 3 \\ 6 \\ 3 \end{bmatrix}.$$

Choose $w_2 = (\frac{1}{3})v_2 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 50 & 0 \\ 0 & 6 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.5: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 1 & 7 \\ -4 & -7 \\ 0 & -4 \\ 1 & 1 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 1 \\ -4 \\ 0 \\ 1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 1 \\ -4 \\ 0 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (2).$$

Subtract all of those components:

$$v_2 = x_2 - (2)w_1 = \begin{bmatrix} 5 \\ 1 \\ -4 \\ -1 \end{bmatrix}.$$

Choose $w_2 = v_2 = \begin{bmatrix} 5 \\ 1 \\ -4 \\ -1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 18 & 0 \\ 0 & 43 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.6: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 3 & -5 \\ -1 & 9 \\ 2 & -9 \\ -1 & 3 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 3 \\ -1 \\ 2 \\ -1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 3 \\ -1 \\ 2 \\ -1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (-3).$$

Subtract all of those components:

$$v_2 = x_2 - (-3)w_1 = \begin{bmatrix} 4 \\ 6 \\ -3 \\ 0 \end{bmatrix}.$$

Choose $w_2 = v_2 = \begin{bmatrix} 4 \\ 6 \\ -3 \\ 0 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 15 & 0 \\ 0 & 61 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.7: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 2 & 4 \\ -5 & -1 \\ 1 & 2 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (\frac{1}{2}).$$

Subtract all of those components:

$$v_2 = x_2 - (\frac{1}{2})w_1 = \begin{bmatrix} 3 \\ \frac{3}{2} \\ \frac{3}{2} \end{bmatrix}.$$

Choose $w_2 = (\frac{2}{3})v_2 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 30 & 0 \\ 0 & 6 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Normalize each column by its own positive length.

$$Q = \begin{bmatrix} \frac{w_1}{\|w_1\|} & \cdots & \frac{w_p}{\|w_p\|} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{30}}{15} & \frac{\sqrt{6}}{3} \\ -\frac{\sqrt{30}}{6} & \frac{\sqrt{6}}{6} \\ \frac{\sqrt{30}}{30} & \frac{\sqrt{6}}{6} \end{bmatrix}.$$

The exact check $Q^T Q = I$ verifies all unit lengths and mutual orthogonality.

Conclusion. The columns of Q are an orthonormal basis for the original column span.

Exercise 6.4.8: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 3 & -3 \\ -4 & 14 \\ 5 & -7 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 3 \\ -4 \\ 5 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 3 \\ -4 \\ 5 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (-2).$$

Subtract all of those components:

$$v_2 = x_2 - (-2)w_1 = \begin{bmatrix} 3 \\ 6 \\ 3 \end{bmatrix}.$$

Choose $w_2 = (\frac{1}{3})v_2 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 50 & 0 \\ 0 & 6 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Normalize each column by its own positive length.

$$Q = \begin{bmatrix} \dfrac{w_1}{\|w_1\|} & \cdots & \dfrac{w_p}{\|w_p\|} \end{bmatrix} = \begin{bmatrix} \frac{3\sqrt{2}}{10} & \frac{\sqrt{6}}{6} \\ -\frac{2\sqrt{2}}{5} & \frac{\sqrt{6}}{3} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{6}}{6} \end{bmatrix}.$$

The exact check $Q^T Q = I$ verifies all unit lengths and mutual orthogonality.

Conclusion. The columns of Q are an orthonormal basis for the original column span.

Exercise 6.4.9: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 3 & -5 & 1 \\ 1 & 1 & 1 \\ -1 & 5 & -2 \\ 3 & -7 & 8 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 3 \\ 1 \\ -1 \\ 3 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 3 \\ 1 \\ -1 \\ 3 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (-2).$$

Subtract all of those components:

$$v_2 = x_2 - (-2)w_1 = \begin{bmatrix} 1 \\ 3 \\ 3 \\ -1 \end{bmatrix}.$$

Choose $w_2 = v_2 = \begin{bmatrix} 1 \\ 3 \\ 3 \\ -1 \end{bmatrix}$.

Column 3.

The component coefficients along the previous w_j are

$$\frac{x_3 \cdot w_j}{w_j \cdot w_j} : \quad (\frac{3}{2}, -\frac{1}{2}).$$

Subtract all of those components:

$$v_3 = x_3 - (\frac{3}{2})w_1 - (-\frac{1}{2})w_2 = \begin{bmatrix} -3 \\ 1 \\ 1 \\ 3 \end{bmatrix}.$$

Choose $w_3 = v_3 = \begin{bmatrix} -3 \\ 1 \\ 1 \\ 3 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 20 & 0 & 0 \\ 0 & 20 & 0 \\ 0 & 0 & 20 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.10: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} -1 & 6 & 6 \\ 3 & -8 & 3 \\ 1 & -2 & 6 \\ 1 & -4 & -3 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} -1 \\ 3 \\ 1 \\ 1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} -1 \\ 3 \\ 1 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (-3).$$

Subtract all of those components:

$$v_2 = x_2 - (-3)w_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \\ -1 \end{bmatrix}.$$

Choose $w_2 = v_2 = \begin{bmatrix} 3 \\ 1 \\ 1 \\ -1 \end{bmatrix}$.

Column 3.

The component coefficients along the previous w_j are

$$\frac{x_3 \cdot w_j}{w_j \cdot w_j} : \quad (\frac{1}{2}, \frac{5}{2}).$$

Subtract all of those components:

$$v_3 = x_3 - (\frac{1}{2})w_1 - (\frac{5}{2})w_2 = \begin{bmatrix} -1 \\ -1 \\ 3 \\ -1 \end{bmatrix}.$$

Choose $w_3 = v_3 = \begin{bmatrix} -1 \\ -1 \\ 3 \\ -1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 12 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 12 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.11: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 1 & 2 & 5 \\ -1 & 1 & -4 \\ -1 & 4 & -3 \\ 1 & -4 & 7 \\ 1 & 2 & 1 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j}: \quad (-1).$$

Subtract all of those components:

$$v_2 = x_2 - (-1)w_1 = \begin{bmatrix} 3 \\ 0 \\ 3 \\ -3 \\ 3 \end{bmatrix}.$$

Choose $w_2 = (\frac{1}{3})v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \\ 1 \end{bmatrix}$.

Column 3.

The component coefficients along the previous w_j are

$$\frac{x_3 \cdot w_j}{w_j \cdot w_j}: \quad (4, -1).$$

Subtract all of those components:

$$v_3 = x_3 - (4)w_1 - (-1)w_2 = \begin{bmatrix} 2 \\ 0 \\ 2 \\ 2 \\ -2 \end{bmatrix}.$$

Choose $w_3 = (\frac{1}{2})v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ -1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.12: Gram–Schmidt with every component removed

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 1 & 3 & 5 \\ -1 & -3 & 1 \\ 0 & 2 & 3 \\ 1 & 5 & 2 \\ 1 & 5 & 8 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \\ 1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (4).$$

Subtract all of those components:

$$v_2 = x_2 - (4)w_1 = \begin{bmatrix} -1 \\ 1 \\ 2 \\ 1 \\ 1 \end{bmatrix}.$$

Choose $w_2 = v_2 = \begin{bmatrix} -1 \\ 1 \\ 2 \\ 1 \\ 1 \end{bmatrix}$.

Column 3.

The component coefficients along the previous w_j are

$$\frac{x_3 \cdot w_j}{w_j \cdot w_j}: \quad (\frac{7}{2}, \frac{3}{2}).$$

Subtract all of those components:

$$v_3 = x_3 - (\frac{7}{2})w_1 - (\frac{3}{2})w_2 = \begin{bmatrix} 3 \\ 3 \\ 0 \\ -3 \\ 3 \end{bmatrix}.$$

Choose $w_3 = (\frac{1}{3})v_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 4 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.13: Gram–Schmidt with every component removed

Use the known orthonormal factor to isolate R.

The supplied Q satisfies $Q^T Q = I_2$. Left multiplication of $A = QR$ by Q^T gives $R = Q^T A$, so

$$R = \begin{bmatrix} 6 & 12 \\ 0 & 6 \end{bmatrix}.$$

Verify the matrix product, not just triangularity.

$$Q = \begin{bmatrix} \frac{5}{6} & -\frac{1}{6} \\ \frac{1}{6} & \frac{5}{6} \\ -\frac{1}{2} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{2} \end{bmatrix}, \qquad QR = \begin{bmatrix} 5 & 9 \\ 1 & 7 \\ -3 & -5 \\ 1 & 5 \end{bmatrix} = A.$$

Conclusion. The displayed upper triangular R is the required factor.

Exercise 6.4.14: Gram–Schmidt with every component removed

Use the known orthonormal factor to isolate R.

The supplied Q satisfies $Q^TQ = I_2$. Left multiplication of $A = QR$ by Q^T gives $R = Q^TA$, so

$$R = \begin{bmatrix} 7 & 7 \\ 0 & 7 \end{bmatrix}.$$

Verify the matrix product, not just triangularity.

$$Q = \begin{bmatrix} -\frac{2}{7} & \frac{5}{7} \\ \frac{5}{7} & \frac{2}{7} \\ \frac{2}{7} & -\frac{4}{7} \\ \frac{4}{7} & \frac{2}{7} \end{bmatrix}, \qquad QR = \begin{bmatrix} -2 & 3 \\ 5 & 7 \\ 2 & -2 \\ 4 & 6 \end{bmatrix} = A.$$

Conclusion. The displayed upper triangular R is the required factor.

Exercise 6.4.15: An exact QR factorization and its checks

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 1 & 2 & 5 \\ -1 & 1 & -4 \\ -1 & 4 & -3 \\ 1 & -4 & 7 \\ 1 & 2 & 1 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j}: \quad (-1).$$

Subtract all of those components:

$$v_2 = x_2 - (-1)w_1 = \begin{bmatrix} 3 \\ 0 \\ 3 \\ -3 \\ 3 \end{bmatrix}.$$

Choose $w_2 = (\frac{1}{3})v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \\ 1 \end{bmatrix}$.

Column 3.

The component coefficients along the previous w_j are

$$\frac{x_3 \cdot w_j}{w_j \cdot w_j}: \quad (4, -1).$$

Subtract all of those components:

$$v_3 = x_3 - (4)w_1 - (-1)w_2 = \begin{bmatrix} 2 \\ 0 \\ 2 \\ 2 \\ -2 \end{bmatrix}.$$

Choose $w_3 = (\frac{1}{2})v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ -1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Normalize each column by its own positive length.

$$Q = \begin{bmatrix} \frac{w_1}{\|w_1\|} & \cdots & \frac{w_p}{\|w_p\|} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{5}}{5} & \frac{1}{2} & \frac{1}{2} \\ -\frac{\sqrt{5}}{5} & 0 & 0 \\ -\frac{\sqrt{5}}{5} & \frac{1}{2} & \frac{1}{2} \\ \frac{\sqrt{5}}{5} & -\frac{1}{2} & \frac{1}{2} \\ \frac{\sqrt{5}}{5} & \frac{1}{2} & -\frac{1}{2} \end{bmatrix}.$$

The exact check $Q^TQ = I$ verifies all unit lengths and mutual orthogonality.

Compute the upper triangular factor from inner products.

$$R = Q^TA = \begin{bmatrix} \sqrt{5} & -\sqrt{5} & 4\sqrt{5} \\ 0 & 6 & -2 \\ 0 & 0 & 4 \end{bmatrix}.$$

Its entry r_{ij} is $q_i \cdot x_j$. For $i > j$, q_i is perpendicular to the earlier-column span containing x_j, so these entries are zero. Direct multiplication gives $QR = A$, verifying the full factorization.

Conclusion. The displayed Q, R give a QR factorization with positive diagonal entries of R.

Exercise 6.4.16: An exact QR factorization and its checks

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} 1 & 3 & 5 \\ -1 & -3 & 1 \\ 0 & 2 & 3 \\ 1 & 5 & 2 \\ 1 & 5 & 8 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \\ 1 \end{bmatrix}$. Choose $w_1 = v_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (4).$$

Subtract all of those components:

$$v_2 = x_2 - (4)w_1 = \begin{bmatrix} -1 \\ 1 \\ 2 \\ 1 \\ 1 \end{bmatrix}.$$

Choose $w_2 = v_2 = \begin{bmatrix} -1 \\ 1 \\ 2 \\ 1 \\ 1 \end{bmatrix}$.

Column 3.

The component coefficients along the previous w_j are

$$\frac{x_3 \cdot w_j}{w_j \cdot w_j} : \quad (\frac{7}{2}, \frac{3}{2}).$$

Subtract all of those components:

$$v_3 = x_3 - (\frac{7}{2})w_1 - (\frac{3}{2})w_2 = \begin{bmatrix} 3 \\ 3 \\ 0 \\ -3 \\ 3 \end{bmatrix}.$$

Choose $w_3 = (\frac{1}{3})v_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 4 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Normalize each column by its own positive length.

$$Q = \begin{bmatrix} \frac{w_1}{\|w_1\|} & \cdots & \frac{w_p}{\|w_p\|} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{2}}{4} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{2}}{4} & \frac{1}{2} \\ 0 & \frac{\sqrt{2}}{2} & 0 \\ \frac{1}{2} & \frac{\sqrt{2}}{4} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{2}}{4} & \frac{1}{2} \end{bmatrix}.$$

The exact check $Q^T Q = I$ verifies all unit lengths and mutual orthogonality.

Compute the upper triangular factor from inner products.

$$R = Q^T A = \begin{bmatrix} 2 & 8 & 7 \\ 0 & 2\sqrt{2} & 3\sqrt{2} \\ 0 & 0 & 6 \end{bmatrix}.$$

Its entry r_{ij} is $q_i \cdot x_j$. For $i > j$, q_i is perpendicular to the earlier-column span containing x_j, so these entries are zero. Direct multiplication gives $QR = A$, verifying the full factorization.

Conclusion. The displayed Q, R give a QR factorization with positive diagonal entries of R.

Exercise 6.4.17: Gram–Schmidt with every component removed

(a) False as stated: $c = 0$ replaces a nonzero basis vector by zero, which makes the family dependent and unable to retain its old span. For $c \neq 0$ it would remain an orthogonal basis.

(b) True. Each residual is obtained by subtracting a combination of earlier directions, and the reverse equation reconstructs the new input. Induction proves equality of each pair of successive spans.

(c) True. Since $Q^T Q = I$, left multiplication of $A = QR$ gives $Q^T A = Q^T QR = R$.

Exercise 6.4.18: Gram–Schmidt with every component removed

(a) False without the requirement that all v_i are nonzero. In a three-dimensional W, the set $\{e_1, e_2, 0\}$ is orthogonal but is not a basis. If the three orthogonal vectors were all nonzero, they would be independent and would form a basis.

(b) True. If $x - \text{proj}_W x = 0$, then $x = \text{proj}_W x \in W$, contradicting $x \notin W$.

(c) True. Gram–Schmidt produces orthonormal columns of Q with exactly the same column span as A; they therefore form a basis of that column space.

Exercise 6.4.19: Gram–Schmidt with every component removed

Suppose $Rx = 0$. Multiply by Q to get $Ax = QRx = 0$. Independence of the columns of A means its homogeneous system has only the zero solution, so $x = 0$. Thus the square $n \times n$ matrix R has trivial null space. By the Invertible Matrix Theorem, it is invertible.

Conclusion. The conclusion requires neither guessing a determinant nor assuming R is already invertible; it follows from independence of A.

Exercise 6.4.20: Gram–Schmidt with every component removed

If $y \in \text{Col}\, A$, write $y = Ax = Q(Rx)$. This represents y as a combination of columns of Q, proving $\text{Col}\, A \subseteq \text{Col}\, Q$. Conversely, if $y \in \text{Col}\, Q$, write $y = Qz$ and set $x = R^{-1}z$. Then $Ax = QR(R^{-1}z) = Qz = y$, so $y \in \text{Col}\, A$.

Conclusion. The two inclusions give $\text{Col}\, A = \text{Col}\, Q$. Invertibility of R is needed for the reverse inclusion.

Exercise 6.4.21: Gram–Schmidt with every component removed

Start with the thin factorization $A = QR$, where Q has n orthonormal columns in $\mathbb{R}^m$ and R is invertible upper triangular. Extend those columns to an orthonormal basis of $\mathbb{R}^m$ by taking an orthonormal basis $Q_\perp$ of $(\operatorname{Col} Q)^\perp$; for example append independent standard coordinate directions and continue Gram–Schmidt. Set $Q_1 = [Q\ Q_\perp]$. Its m columns are orthonormal, so it is an orthogonal $m \times m$ matrix. Then

$$Q_1 \begin{bmatrix} R \\ 0_{(m-n)\times n} \end{bmatrix} = QR + Q_\perp 0 = A.$$

Conclusion. Adjoining the orthonormal complement produces the full QR factorization. The triangular factor in the top block remains the same R.

Exercise 6.4.22: Gram–Schmidt with every component removed

Using the orthogonal projection formula and nonzero basis vectors,

$$T(x) = \sum_{j=1}^{p} \frac{x \cdot u_j}{u_j \cdot u_j} u_j.$$

For any x, y and scalars a, b, substitute $ax+by$ and distribute each numerator. The fixed denominators give

$$T(ax + by) = \sum_j \frac{a(x \cdot u_j) + b(y \cdot u_j)}{u_j \cdot u_j} u_j = aT(x) + bT(y).$$

Conclusion. Projection onto a fixed subspace is linear.

Exercise 6.4.23: Gram–Schmidt with every component removed

Partition $Q = [Q_1\ Q_2]$ after its first p columns and partition the upper triangular matrix R compatibly:

$$R = \begin{bmatrix} R_{11} & R_{12} \\ 0 & R_{22} \end{bmatrix}.$$

The first p columns of QR are therefore $Q_1 R_{11}$, so $A_1 = Q_1 R_{11}$. A subset of the orthonormal columns of Q remains orthonormal, and R_{11} is upper triangular. Its diagonal entries are the first p nonzero diagonal entries of R, so it is invertible.

Conclusion. Retain the first p columns of Q and the leading $p \times p$ block of R to obtain a QR factorization of A_1.

Exercise 6.4.24: Four checked orthogonal directions in five coordinates

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} -10 & 13 & 7 & -11 \\ 2 & 1 & -5 & 3 \\ -6 & 3 & 13 & -3 \\ 16 & -16 & -2 & 5 \\ 2 & 1 & -5 & -7 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} -10 \\ 2 \\ -6 \\ 16 \\ 2 \end{bmatrix}$. Choose $w_1 = (\frac{1}{2})v_1 = \begin{bmatrix} -5 \\ 1 \\ -3 \\ 8 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (-2).$$

Subtract all of those components:

$$v_2 = x_2 - (-2)w_1 = \begin{bmatrix} 3 \\ 3 \\ -3 \\ 0 \\ 3 \end{bmatrix}.$$

Choose $w_2 = (\frac{1}{3})v_2 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \\ 1 \end{bmatrix}$.

Column 3.

The component coefficients along the previous w_j are

$$\frac{x_3 \cdot w_j}{w_j \cdot w_j} : \quad (-1, -4).$$

Subtract all of those components:

$$v_3 = x_3 - (-1)w_1 - (-4)w_2 = \begin{bmatrix} 6 \\ 0 \\ 6 \\ 6 \\ 0 \end{bmatrix}.$$

Choose $w_3 = (\frac{1}{6})v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$.

Column 4.

The component coefficients along the previous w_j are

$$\frac{x_4 \cdot w_j}{w_j \cdot w_j} : \quad (1, -3, -3).$$

Subtract all of those components:

$$v_4 = x_4 - (1)w_1 - (-3)w_2 - (-3)w_3 = \begin{bmatrix} 0 \\ 5 \\ 0 \\ 0 \\ -5 \end{bmatrix}.$$

Choose $w_4 = (\frac{1}{5})v_4 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 100 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Conclusion. The vectors $w_1, \ldots, w_p$ are an orthogonal basis for the original column span.

Exercise 6.4.25: QR factors for the five-by-four matrix

Process the original columns in their given order.

$$A = [x_1 \ \cdots \ x_p] = \begin{bmatrix} -10 & 13 & 7 & -11 \\ 2 & 1 & -5 & 3 \\ -6 & 3 & 13 & -3 \\ 16 & -16 & -2 & 5 \\ 2 & 1 & -5 & -7 \end{bmatrix}.$$

To keep the arithmetic readable, a nonzero intermediate vector may be multiplied by a positive scalar to clear fractions or common factors. This does not change its direction or any later projection onto it. Denote these convenient orthogonal vectors by w_j.

Column 1.

Start with $v_1 = x_1 = \begin{bmatrix} -10 \\ 2 \\ -6 \\ 16 \\ 2 \end{bmatrix}$. Choose $w_1 = (\frac{1}{2})v_1 = \begin{bmatrix} -5 \\ 1 \\ -3 \\ 8 \\ 1 \end{bmatrix}$.

Column 2.

The component coefficients along the previous w_j are

$$\frac{x_2 \cdot w_j}{w_j \cdot w_j} : \quad (-2).$$

Subtract all of those components:

$$v_2 = x_2 - (-2)w_1 = \begin{bmatrix} 3 \\ 3 \\ -3 \\ 0 \\ 3 \end{bmatrix}.$$

Choose $w_2 = (\frac{1}{3})v_2 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \\ 1 \end{bmatrix}$.

Column 3.

The component coefficients along the previous w_j are

$$\frac{x_3 \cdot w_j}{w_j \cdot w_j} : \quad (-1, -4).$$

Subtract all of those components:

$$v_3 = x_3 - (-1)w_1 - (-4)w_2 = \begin{bmatrix} 6 \\ 0 \\ 6 \\ 6 \\ 0 \end{bmatrix}.$$

Choose $w_3 = (\frac{1}{6})v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$.

Column 4.

The component coefficients along the previous w_j are

$$\frac{x_4 \cdot w_j}{w_j \cdot w_j}: \quad (1, -3, -3).$$

Subtract all of those components:

$$v_4 = x_4 - (1)w_1 - (-3)w_2 - (-3)w_3 = \begin{bmatrix} 0 \\ 5 \\ 0 \\ 0 \\ -5 \end{bmatrix}.$$

Choose $w_4 = (\frac{1}{5})v_4 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}$.

Verify orthogonality and equality of spans.

Set $V = [w_1 \ \cdots \ w_p]$. Then

$$V^T V = \begin{bmatrix} 100 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}.$$

The diagonal is positive and all off-diagonal entries are zero, so the w_j are nonzero and orthogonal, hence independent. Each new w_k is a nonzero multiple of x_k minus a combination of earlier vectors. Conversely, that equation expresses x_k using w_k and the earlier vectors. Induction gives equal spans at every stage.

Normalize each column by its own positive length.

$$Q = \begin{bmatrix} \frac{w_1}{\|w_1\|} & \cdots & \frac{w_p}{\|w_p\|} \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{\sqrt{3}}{3} & 0 \\ \frac{1}{10} & \frac{1}{2} & 0 & \frac{\sqrt{2}}{2} \\ -\frac{3}{10} & -\frac{1}{2} & \frac{\sqrt{3}}{3} & 0 \\ \frac{4}{5} & 0 & \frac{\sqrt{3}}{3} & 0 \\ \frac{1}{10} & \frac{1}{2} & 0 & -\frac{\sqrt{2}}{2} \end{bmatrix}.$$

The exact check $Q^T Q = I$ verifies all unit lengths and mutual orthogonality.

Compute the upper triangular factor from inner products.

$$R = Q^T A = \begin{bmatrix} 20 & -20 & -10 & 10 \\ 0 & 6 & -8 & -6 \\ 0 & 0 & 6\sqrt{3} & -3\sqrt{3} \\ 0 & 0 & 0 & 5\sqrt{2} \end{bmatrix}.$$

Its entry r_{ij} is $q_i \cdot x_j$. For $i > j$, q_i is perpendicular to the earlier-column span containing x_j, so these entries are zero. Direct multiplication gives $QR = A$, verifying the full factorization.

Conclusion. The displayed Q, R give a QR factorization with positive diagonal entries of R.

Exercise 6.4.26: Gram–Schmidt with every component removed

Implement the specified orthonormal update in exact arithmetic.

```
from sympy import Matrix, zeros, sqrt, simplify
A = Matrix([[-10,13,7,-11],[2,1,-5,3],[-6,3,13,-3],
            [16,-16,-2,5],[2,1,-5,-7]])
Q = zeros(A.rows,0)
for k in range(A.cols):
    x = A[:,k]
    v = simplify(x-Q*(Q.T*x))
    q = simplify(v/sqrt(v.dot(v)))
    Q = Q.row_join(q)
R = simplify(Q.T*A)
assert simplify(Q.T*Q) == Matrix.eye(4)
assert simplify(Q*R) == A
print(Q)
print(R)
```

Record the actual computed factors.

$$Q = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{\sqrt{3}}{3} & 0 \\ \frac{1}{10} & \frac{1}{2} & 0 & \frac{\sqrt{2}}{2} \\ -\frac{3}{10} & -\frac{1}{2} & \frac{\sqrt{3}}{3} & 0 \\ \frac{4}{5} & 0 & \frac{\sqrt{3}}{3} & 0 \\ \frac{1}{10} & \frac{1}{2} & 0 & -\frac{\sqrt{2}}{2} \end{bmatrix}, \qquad R = \begin{bmatrix} 20 & -20 & -10 & 10 \\ 0 & 6 & -8 & -6 \\ 0 & 0 & 6\sqrt{3} & -3\sqrt{3} \\ 0 & 0 & 0 & 5\sqrt{2} \end{bmatrix}.$$

The update subtracts all previous components at once through $Q(Q^T x)$. Nonzero residuals follow from independent input columns. These exact factors agree with Exercise 25; the two displayed assertions verify unit orthogonality and reconstruction.

Conclusion. The commands implement the requested recurrence and produce the explicit checked QR factorization above.